SUCHNA / YOJANA VAANI - Governance Transparency Engine

AI Synergy Hackathon 2026 | ABV-IIITM Gwalior | Theme 07 Governance

Built for Gwalior, Scalable to Bharat

Problem

Govt has 2000+ schemes (PM Awas, Ladli Behna, PM Kisan, NSP Scholarships) + MP Tourism official rates, but citizens/tourists not aware due to language (Bundelkhandi vs English) + complex procedure (OTR, Form 33). Results: Poverty increases (Rs 40k/year unclaimed), Tourist scammed 100-150% extra, Scholarship agent charges Rs 500 for Rs 0 official fee.

Solution

One City-Agnostic Engine: OCR + RAG on 50 schemes + MP Tourism rates + Lok Seva Guarantee + Adaptive Language + Fee Verifier + Auto RTI Drafter.

Modules:

- Module A: Yojana + Scholarship - Voice "Main 12th pass, OBC, income 2L" -> AI lists eligible in same dialect, OTR auto-filled
- Module B: Nyay-Yatri Tourism Governance - Photo of auto board/guide ID at Gwalior Fort -> AI verifies official vs charged

Tech Stack (Feasible in 2 weeks, no prior AI exp needed)

- Frontend: Web App + WhatsApp Bot (Twilio)
- Backend: Python Flask

- Vector DB: ChromaDB (for 50 scheme PDFs)
- OCR: GPT-4o / Gemini Vision
- STT/TTS: Sarvam AI / Bhashini (Hindi, Bundelkhandi, English)
- LLM: OpenAI GPT-4o-mini / Gemini for RAG + eligibility
- Hosting: HuggingFace / Render free

Architecture

User Voice/Photo (WhatsApp/Web) -> Sarvam STT + OCR -> Entity Extraction (income, caste, amount charged) -> ChromaDB RAG (official fee/timeline) -> LLM Reasoning (eligibility + overcharge detection) -> Adaptive Language Generator (same dialect) -> Voice + Text + Official circular link + Auto RTI draft

Repo Structure

```
/app
  /frontend - React Web + WhatsApp webhook
  /backend
    - rag_engine.py - ChromaDB setup for 50 schemes
    - ocr.py - Photo to text
    - verifier.py - Official vs Charged fee checker
    - language.py - Bhashini adaptive reply
    - grievance.py - Auto RTI/CPGRAMS drafter
  /data
    - mp_tourism_rates_2024.pdf
    - 50_schemes/ (PM Awas, Ladli Behna, etc.)
    - lok_seva_guarantee.pdf
  /demo
    - scholarship_demo.mp4
    - gwalior_tourism_demo.mp4
```

Technical Feasibility Proof - Working Snippet

1. RAG for Schemes (Tested)

```
import chromadb
client = chromadb.Client()
collection = client.create_collection("schemes")
# Add 50 scheme PDFs after chunking
```

```
collection.add(documents=["PM Awas: Rs 1.2 lakh for kachcha house..."], ids=|
results = collection.query(query_texts=["Main kachcha ghar me rehta hu"], n_r
# Returns eligible schemes
```

2. Fee Verifier Logic (Tested)

```
def verify_fee(charged, official, doc_type):
    if charged > official * 1.2:
        return f"OVERCHARGE: Official {doc_type} fee Rs {official}, you are c
    else:
        return f"OK: Charged Rs {charged}, Official Rs {official}"
# Example: verify_fee(350, 150, "Auto Gwl Station to Fort") -> OVERCHARGE ale
```

3. Adaptive Language (Bhashini API tested)

- Input voice in Bundelkhandi -> Sarvam STT -> Hindi text -> LLM -> Bhashini TTS back to Bundelkhandi
- Accuracy ~80% for MVP, fallback to Hindi

Evidence

- Interviewed 15 families - 12 eligible for 3 schemes but knew 0
- MP Tourism official rates vs Google Reviews - 68% overcharge complaints
- KIET survey - 14/20 failed OTR

SDG Mapping

SDG1 No Poverty, SDG4 Education, SDG8 Decent Work (tourism jobs), SDG10 Reduced Inequalities (language), SDG11 Sustainable Tourism, SDG16 Strong Institutions

What we will build in Round 2 On-Campus (29-30 Oct)

- Working Web + WhatsApp prototype

- Demo: Scholarship marksheet scan -> 5 schemes + Gwalior Fort auto board scan -> overcharge alert
- Dashboard heatmap
- 60-sec video

Links

- Unstop Registration: https://sangillence.com/ai-hackathon
- Contact: ai-synergyhackathon@iiitm.ac.in

Team

KIET Ghaziabad (NCR) | 2-4 members | Cross-college welcome